

CHAPTER 1

INTRODUCTION

1.1 Introduction

EduGenie is an AI-powered educational learning assistant developed using Google Gemini, LaMini-Flan-T5, FastAPI, HTML, and CSS. The application helps students by answering questions, explaining concepts, generating quizzes, summarizing educational content, and recommending learning paths. It combines modern Generative AI with a simple web interface to create an interactive learning experience. The modular design allows each feature to operate independently while sharing a common backend. This improves maintainability and scalability. EduGenie demonstrates how artificial intelligence can simplify learning by providing instant academic support anytime and anywhere.

The project is intended for students, teachers, and self-learners who need reliable educational assistance. By integrating AI models with a lightweight FastAPI backend, EduGenie delivers fast responses and an easy-to-use interface.

1.2 Background

Digital learning has become an important part of education. Students often switch between multiple websites for explanations, quizzes, notes, and study plans. Generative AI can bring these services together in one intelligent application. EduGenie addresses this need by providing multiple educational services through a single platform, reducing learning time and improving productivity.

1.3 Problem Statement

Students frequently struggle to find accurate explanations, generate practice questions, summarize long topics, and organize learning resources. Existing solutions are fragmented and often lack personalization. EduGenie solves these problems by integrating AI-powered educational tools into one web application.

1.4 Objectives

- Develop an AI-powered learning assistant.
- Answer academic questions accurately.
- Explain difficult concepts in simple language.
- Generate quizzes for practice.
- Summarize lengthy educational content.
- Recommend personalized learning paths.
- Provide a simple and user-friendly web interface.

1.5 Scope

EduGenie supports question answering, concept explanation, quiz generation, text summarization, and learning recommendations through a FastAPI backend and HTML/CSS frontend. The architecture also allows future integration with databases, authentication, multilingual support, and cloud deployment.

1.6 Existing System

Traditional learning requires students to use different platforms for different tasks. This consumes time and provides inconsistent learning experiences. Many systems do not offer personalized AI assistance.

1.7 Proposed System

EduGenie integrates all major educational services into one AI-powered application. Users interact through a web interface while the backend communicates with Google Gemini and LaMini-Flan-T5 to generate intelligent responses. The modular architecture improves scalability, maintainability, and ease of future enhancement.

1.8 Features

- Question Answering
- Explanation Module
- Quiz Generation
- Summary Module
- Learning Path Recommendation
- FastAPI Backend
- HTML/CSS Frontend
- Google Gemini Integration

1.9 Applications

Educational institutions, self-learning platforms, competitive exam preparation, online tutoring, and classroom support.

1.10 Chapter Summary

This chapter introduced the EduGenie project, its objectives, background, scope, existing and proposed systems, and major features. The following chapters describe the requirements, system design, implementation, testing, deployment, and conclusions in detail.